MODERN PHYSICS

JEE MAIN + ADVANCED · FORMULA REVISION SHEET

Dual nature · Photoelectric · X-rays · Atoms
Nuclei · Radioactivity · Nuclear reactions
★ = standard JEE result filled from syllabus gaps

Boxed gold bar = must-remember relation Navy bar = definition / SI form Conditions in amber Source PDF is primary; signs/ε_0/$\hbar$ typos in the source are corrected here

0 · CONSTANTS & CONVERSIONS

Quantity	JEE working value
Planck h	$6.626\times10^{-34}\,\text{J s} = 4.136\times10^{-15}\,\text{eV s}$
$\hbar = h/2\pi$	$1.055\times10^{-34}\,\text{J s}$
hc	$12400\,\text{eV}\cdot\text{Å} = 1240\,\text{eV}\cdot\text{nm}$
$c,\ e,\ m_e$	$3\times10^{8}\,\text{m/s};\ 1.6\times10^{-19}\,\text{C};\ 9.1\times10^{-31}\,\text{kg}$
Rydberg R_∞	$1.097373\times10^{7}\,\text{m}^{-1}$
Bohr radius a_0	$0.529\,\text{Å} = 5.29\times10^{-11}\,\text{m}$
$1\,\text{eV},\ 1\,\text{u}$	$1.6\times10^{-19}\,\text{J};\ 931.5\,\text{MeV}/c^2 = 1.6605\times10^{-27}\,\text{kg}$
Fine structure α ★	$\alpha = e^2/(4\pi\varepsilon_0\hbar c)\approx 1/137$
Bohr magneton μ_B ★	$e\hbar/2m_e = 9.27\times10^{-24}\,\text{J/T}$
Compton λ_C ★	$h/(m_e c) = 2.43\times10^{-12}\,\text{m} = 0.0243\,\text{Å}$
$k = 1/(4\pi\varepsilon_0)$	$9\times10^{9}\,\text{N m}^2\text{C}^{-2}$

Workhorse: photon energy $E(\text{eV}) = 12400/\lambda(\text{Å}) = 1240/\lambda(\text{nm})$.

1 · DUAL NATURE OF RADIATION & MATTER

1.1 Photons ★

$$E = h\nu = \frac{hc}{\lambda},\qquad p = \frac{h}{\lambda} = \frac{E}{c},\qquad m_{\text{rel}} = \frac{E}{c^2}$$

Number of photons in time t from power P: $N = Pt/(h\nu) = P\lambda t/(hc)$.
Intensity $I = (\text{energy/area/time})$.

A photon interacts with **one** electron (Einstein). Intensity ∝ number of photons, not energy of one photon.

1.2 Photoelectric effect

Emission of electrons from a metal by light of frequency $\geq \nu_0$ (or $\lambda \leq \lambda_0$). Instantaneous — no measurable time lag.

Einstein's photoelectric equation

$$h\nu = \phi_0 + K_{\max} = h\nu_0 + \tfrac{1}{2}mv_{\max}^2$$
$$K_{\max} = eV_s = h(\nu - \nu_0)$$
$$K_{\max} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

$\phi_0 = h\nu_0 = hc/\lambda_0$ = work function. V_s = stopping potential (retarding voltage that makes photocurrent zero).

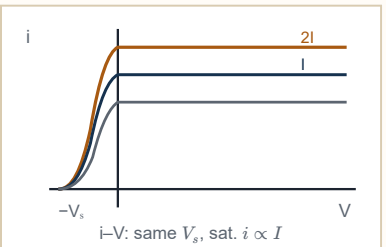
Emission only if $\nu \geq \nu_0$. Raising intensity cannot start emission below threshold.

Change	Current i	$K_{\max}$, V_s
Intensity ↑ (ν fixed $> \nu_0$)	sat. $i \propto I$	unchanged
Frequency ↑ (I fixed)	sat. i same	both ↑
Different metal	—	ν_0 changes
Point source distance ★	$I \propto 1/r^2$	—

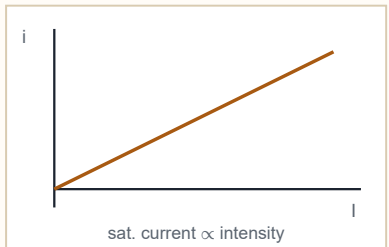
Graphs (JEE Main staples)

$$V_s = \frac{h}{e}\nu - \frac{\phi_0}{e}$$

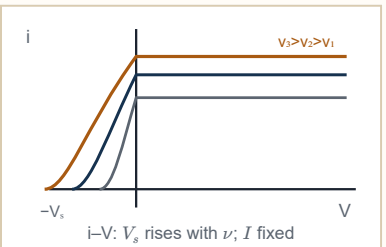
Slope h/e (universal). ν-intercept $= \nu_0$. $K_{\max}$–ν slope $= h$. Parallel lines for different metals.



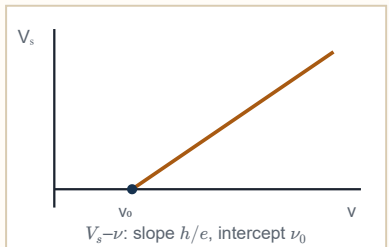
i–V: same V_s, sat. $i \propto I$



sat. current ∝ intensity



i–V: V_s rises with ν; I fixed



V_s–ν: slope h/e, intercept ν_0

1.3 Compton effect ★

Photon–free electron elastic scatter. Wavelength increases; photon loses energy.

$$\Delta\lambda = \lambda' - \lambda = \lambda_C(1-\cos\theta)$$
$$\lambda_C = \frac{h}{m_e c} = 0.0243\,\text{Å}$$

$\theta = 0 \Rightarrow \Delta\lambda = 0;\ \theta = 90^\circ \Rightarrow \Delta\lambda = \lambda_C;\ \theta = 180^\circ \Rightarrow \Delta\lambda = 2\lambda_C$ (max).

$$K_e = h\nu - h\nu' = \frac{h\nu\,\alpha(1-\cos\theta)}{1+\alpha(1-\cos\theta)}$$
$$\alpha = h\nu/(m_e c^2)$$

$\Delta\lambda$ independent of λ and of the scatterer. Recoil electron and photon: coplanar; ϕ (electron) vs θ linked by momentum conservation.

1.4 Pair production / annihilation ★

Photon → $e^- + e^+$ in Coulomb field of a nucleus. Threshold $E_\gamma \geq 2m_e c^2 = 1.022\,\text{MeV}$. Reverse: annihilation yields two $511\,\text{keV}$ photons (to conserve p).

1.5 de Broglie (matter) waves

Every material particle has wavelength $\lambda = h/p$. Observable for light particles (electrons).

$$\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mK}}\qquad (K = \tfrac{1}{2}mv^2 \text{ non-rel.})$$

Charged particle accelerated by V: $K = qV$, so $\lambda = h/\sqrt{2mqV}$.

$$\lambda_e = \frac{12.27}{\sqrt{V}}\ \text{Å}\qquad (V \text{ in volts};\ m_e = 9.1\times10^{-31}\,\text{kg})$$

Source numerical: $h = 6.62\times10^{-34}$. Some keys use $12.25/\sqrt{V}$ or $12.26/\sqrt{V}$. Use the value given in the question.

★ Other species (non-rel., V in volts): $\lambda_p \approx 0.286/\sqrt{V}$ Å, $\lambda_\alpha \approx 0.101/\sqrt{V}$ Å.

Relativistic (when $K \gtrsim mc^2$) ★

$$\lambda = \frac{hc}{\sqrt{K(K+2mc^2)}}$$

From $E_{\text{tot}}^2 = p^2c^2 + m^2c^4$. Use when $K \gtrsim mc^2$.

★ Bohr orbit as a standing wave: $2\pi r_n = n\lambda \Leftrightarrow mvr = n\hbar$ (quantisation).

★ Group velocity = particle velocity v; phase velocity $v_p = E/p = c^2/v > c$ (massive particle).

Davisson–Germer: electron diffraction from Ni; Bragg $2d\sin\theta = n\lambda$. Historic peak: $54\,\text{eV},\ \theta = 50^\circ$.

1.6 Heisenberg uncertainty ★

$$\Delta x\,\Delta p_x \geq \frac{\hbar}{2} = \frac{h}{4\pi},\qquad \Delta E\,\Delta t \geq \frac{\hbar}{2}$$

JEE keys also accept $\geq \hbar$ or $\geq h/4\pi$ if the problem states the form. Estimate ground-state energy of a particle confined to L: $\Delta p \sim h/(4\pi L)$, $K \sim (\Delta p)^2/2m$.

If a photon of λ locates an electron, $\Delta x \sim \lambda$ and $\Delta p \sim h/\lambda$.

2 · X-RAYS

EM waves, $\lambda \sim 0.01$–$1\,\text{nm}$. Coolidge tube: electrons of KE eV strike a metal target.

2.1 Continuous (bremsstrahlung)

Electron decelerated in nuclear field → photons of a continuum of energies up to eV.

$$\lambda_{\min} = \frac{hc}{eV} = \frac{12400}{V}\ \text{Å},\qquad \nu_{\max} = \frac{eV}{h}$$

Duane–Hunt. $\lambda_{\min}$ independent of target Z; decreases as tube voltage ↑.

★ Intensity of continuum ∝ ZV^2 (empirical). Most electrons heat the target, not X-rays.

2.2 Characteristic X-rays

Incident electron knocks out a K (or L, …) electron. A higher-shell electron fills the vacancy → photon of discrete energy (K, L, M series).

Line	Jump	Photon energy
K_α	$2 \to 1$	$10.2(Z-1)^2\,\text{eV}$
K_β	$3 \to 1$	$13.6(Z-1)^2(1-1/9)\,\text{eV}$

L_α	$3 \to 2$	$13.6(Z-7.4)^2(1/4 - 1/9)$ eV ★

$\lambda_{K_\alpha} > \lambda_{K_\beta}$ and $E_{K_\beta} > E_{K_\alpha}$. Characteristic lines appear only if $eV \geq$ ionisation energy of that shell.

2.3 Moseley's law ★

$$\sqrt{\nu} = a(Z-b)$$
$$K_\alpha:\ \nu = \tfrac{3}{4}Rc(Z-1)^2,\ \ b=1$$

$b \approx 7.4$ for L series. $\sqrt{\nu}$ vs Z is linear. Explains ordering of Co/Ni etc.

2.4 Bragg diffraction ★

$$2d\sin\theta = n\lambda \qquad (n = 1, 2, \dots \text{ order})$$



X-ray spectrum: cutoff λ_{min} + characteristic spikes

3 · ATOMS — RUTHERFORD SCATTERING

α on thin Au foil, ZnS scintillations:

- Most α undeflected or deflected $\sim 1^\circ \Rightarrow$ atom mostly empty; +charge in a tiny nucleus.
- A few deflected by $\geq 90^\circ \Rightarrow$ nucleus massive, concentrated charge.

Distance of closest approach (head-on, KE K) ★

$$d = \frac{1}{4\pi\varepsilon_0}\frac{(2e)(Ze)}{K} \qquad (\alpha \text{ on nucleus } Z)$$

At turning point $K = U(d)$. Estimates nuclear size (upper bound).

Impact parameter ★

$$b = \frac{k\,q_1 q_2}{2K}\cot\frac{\theta}{2} = \frac{k\,Ze^2}{K}\cot\frac{\theta}{2}\ (\alpha)$$

$b = 0 \Rightarrow \theta = 180^\circ$; large $b \Rightarrow$ small θ.

★ Scattered number: $N(\theta) \propto \dfrac{ntZ^2}{K^2\sin^4(\theta/2)}$ (Rutherford formula). t = foil thickness, n = number density.

4 · BOHR MODEL (H AND H-LIKE)

Postulates. (1) Electron in circular Coulomb orbit about nucleus; stationary states — no radiation. (2) Radiation only in a jump: $h\nu = |E_i - E_f|$. (3) Angular momentum quantised:

$$L = mvr = n\hbar = \frac{nh}{2\pi}, \quad n = 1, 2, 3, \dots$$

Force balance + quantisation ⇒

$$r_n = \frac{4\pi\varepsilon_0 n^2\hbar^2}{mZe^2} = \frac{n^2 a_0}{Z}$$
$$a_0 = 0.529\ \text{Å}$$
$$v_n = \frac{Ze^2}{4\pi\varepsilon_0 n\hbar} = \frac{Z\alpha c}{n}$$
$$v_1(\text{H}) = 2.18\times10^6\ \text{m/s}$$
$$E_n = -\frac{13.6\,Z^2}{n^2}\ \text{eV}$$

SI energy: $E_n = -mZ^2e^4/(8\varepsilon_0^2 n^2\hbar^2)$. H: $Z=1$; He^+: $Z=2$; Li^{2+}: $Z=3$. Source PDF omitted Z and the minus sign on E_n.

Qty	Relation	∝
KE	$K = -E_n = 13.6Z^2/n^2$ eV	Z^2/n^2
PE	$U = 2E_n = -27.2Z^2/n^2$ eV	Z^2/n^2
Virial	$\lvert U\rvert = 2K$, $E = K + U = -K$	—
r_n	n^2a_0/Z	n^2/Z
v_n	$\alpha c\,Z/n$	Z/n
T_n	$T_1 n^3/Z^2$; $T_1 = 1.52\times10^{-16}$ s	n^3/Z^2
$f = 1/T$	$f_1 Z^2/n^3$	Z^2/n^3
$I = ef$ ★	$ev/(2\pi r)$	Z^2/n^3
$\mu = IA$ ★	$n\mu_B = ne\hbar/(2m)$	n
B at nucleus ★	$\mu_0 I/(2r)$	Z^3/n^5

★ Ionisation energy from level n: $13.6Z^2/n^2$ eV. Excitation $1 \to n$: $13.6Z^2(1-1/n^2)$ eV.

★ Ground-state values (H): $r_1 = a_0$, $v_1 = c/137$, $E_1 = -13.6$ eV, $K = 13.6$ eV, $U = -27.2$ eV.

4.1 Spectra of H / H-like

Photon energy = difference of stationary-state energies (emission: $n_u \to n_l$; absorption: reverse).

$$\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_l^2} - \frac{1}{n_u^2}\right)$$
$$R = 1.097\times10^7\ \text{m}^{-1}$$
$$\Delta E = 13.6\,Z^2\left(\frac{1}{n_l^2} - \frac{1}{n_u^2}\right)\text{eV}$$

Series	n_l	Region	H first line / limit
Lyman	1	UV	1216 Å / 911 Å
Balmer	2	visible	6563 Å / 3646 Å
Paschen	3	IR	18751 Å / 8204 Å
Brackett	4	IR	limit 14584 Å
Pfund	5	IR	limit 22790 Å

Series limit: $1/\lambda = RZ^2/n_l^2$. First line of a series: $n_u = n_l + 1$.

$$N_{\text{lines}}(n_2 \to n_1) = \frac{(n_2-n_1)(n_2-n_1+1)}{2}$$
$$n_2 \to 1:\quad \frac{n_2(n_2-1)}{2}$$

H energy ladder (eV): $n=1: -13.60;\ 2: -3.40;\ 3: -1.51;\ 4: -0.85;\ 5: -0.54;\ \infty: 0$.

Visible Balmer: Hα 6563 Å (red), Hβ 4861, Hγ 4340, Hδ 4102.

4.2 Reduced mass, isotopes, shortcuts ★

$$\mu = \frac{mM}{m+M}, \qquad R_H = R_\infty\frac{\mu}{m} = \frac{R_\infty}{1+m/M}$$

Replace $m \to \mu$ in r_n, E_n. Heavier nucleus ⇒ slightly larger $|E|$ and R. Deuterium lines slightly blue-shifted vs H.

★ For H-like in state n, $Z_{\text{eff}} = Z$. Muonic atom: $m_\mu \approx 207m_e$ ⇒ radii $\sim 1/207$, energies $\times 207$.

★ Photon wavelength from $n_2 \to n_1$: $\lambda = hc/|\Delta E|$. If λ in Å, $\lambda = 12400/\Delta E_{\text{eV}}$.

★ Resonance absorption: incident photon energy must match a Bohr difference (to within natural width).

★ If KE of electron in first Bohr orbit is K, then $E = -K$, $U = -2K$, $v = \sqrt{2K/m}$, $r = ke^2/(2K)$.

5 · ATOMIC NUCLEUS

Size $\sim$ femtometre ($1\,\text{fm} = 10^{-15}$ m). Mass number $A = Z + N$.

$$R = R_0 A^{1/3}, \qquad R_0 \approx 1.2\,\text{fm} = 1.2\times10^{-15}\,\text{m}$$

Some keys use $R_0 = 1.1$ or 1.3 fm — use the given value.

Volume $\propto A$. Nuclear density nearly constant:

$$\rho = \frac{3m_N}{4\pi R_0^3} \approx 2.3\times10^{17}\,\text{kg m}^{-3}$$

★ Surface area $\propto A^{2/3}$. Closest approach of two nuclei $\sim R_1 + R_2 = R_0(A_1^{1/3} + A_2^{1/3})$.

5.1 Mass defect & binding energy ★

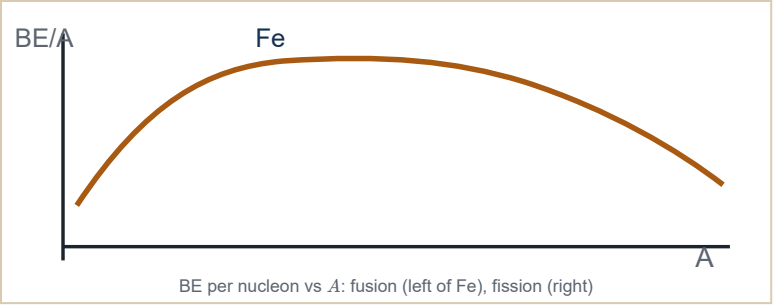
$$\Delta m = Zm_p + (A-Z)m_n - M_{\text{nuc}}$$
$$\text{atomic form: } \Delta m = Zm_H + (A-Z)m_n - M_{\text{atom}}$$
$$\text{BE} = \Delta m\,c^2 = \Delta m(\text{u}) \times 931.5\,\text{MeV}$$

BE per nucleon = BE/A. Peak near ^{56}Fe (≈ 8.8 MeV). Higher BE/A ⇒ more stable.

Packing fraction $f = (M-A)/A$. Negative f ⇔ generally tightly bound.

Useful masses: $m_p = 1.007276$ u, $m_n = 1.008665$ u, $m_H = 1.007825$ u, $m_e = 0.0005486$ u, $\alpha = 4.002603$ u. BE(α) ≈ 28.3 MeV; deuteron ≈ 2.22 MeV.

★ Separation energy of a nucleon ≈ BE difference of parent and residual nucleus.



BE per nucleon vs A: fusion (left of Fe), fission (right)

Nuclear force: short-range ($\sim 2\,\text{fm}$), charge-independent, saturating, spin-dependent, strongest of the three common interactions. Magic numbers ★: $2, 8, 20, 28, 50, 82, 126$.

$N/Z \approx 1$ (light); $N/Z \to \sim 1.6$ (heavy) for β-stability.

5.2 Q-value & nuclear reactions ★

$a + X \to Y + b$. Conserved: A, Z, energy, momentum, lepton number (in β).

$$Q = (m_{\text{initial}} - m_{\text{final}})c^2$$

$Q > 0$ exoergic (possible at low KE); $Q < 0$ endoergic, threshold in lab (target at rest):

$$K_{\text{th}} = -Q\frac{m_a + m_X}{m_X} \approx -Q\left(1 + \frac{m_a}{m_X}\right)$$

Fission: heavy nucleus $\to$ two medium fragments + 2–3 n. $^{235}\text{U} + n \to$ fragments $+ \sim 200\,\text{MeV}$. Fusion: light nuclei combine; $4p \to {}^4\text{He} + 2e^+ + 2\nu + 26.7\,\text{MeV}$ (solar pp chain net).

6 · RADIOACTIVITY

- Statistical: only probabilities; a given nucleus may decay at once or "never".
- Independent of T, P, E/B fields (chemical state too, except tiny electron-capture effects).
- Energy from the nucleus itself. α/β change $Z \Rightarrow$ a new element.

Decay law

$$\frac{dN}{dt} = -\lambda N, \qquad N = N_0 e^{-\lambda t}, \qquad A = \lambda N = A_0 e^{-\lambda t}$$

λ = decay constant (probability of decay per unit time). A = activity.

Units ★: $1\,\text{Bq} = 1$ decay/s; $1\,\text{Ci} = 3.7 \times 10^{10}\,\text{Bq}$.

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}, \qquad \tau = \frac{1}{\lambda} = \frac{T_{1/2}}{0.693} = 1.443\,T_{1/2}$$

$T_{1/2}$: time for $N \to N_0/2$. Mean (average) life τ: average survival time.

$$N = N_0\,2^{-t/T_{1/2}}$$
$$t = \frac{2.303}{\lambda}\log_{10}(N_0/N)$$

After n half-lives: remaining fraction $= 2^{-n}$; decayed $= 1 - 2^{-n}$.

Survival probability of one nucleus in time t: $e^{-\lambda t}$. Decay probability: $1 - e^{-\lambda t}$.

For $t \ll T_{1/2}$: $\approx \lambda t$.

★ Specific activity $= A/m = \lambda N_A/M$ for a pure sample (M = molar mass).

★ $\ln N$ (or $\ln A$) vs t: straight line, slope $= -\lambda$.

6.1 α, β, γ decay ★

Mode	$\Delta A, \Delta Z$	Key relation
α	$-4, -2$	${}^A_Z X \to {}^{A-4}_{Z-2}Y + \alpha$
β^-	$0, +1$	$n \to p + e^- + \bar{\nu}$; $Q = [M_X - M_Y]c^2$
β^+	$0, -1$	$Q = [M_X - M_Y - 2m_e]c^2$
EC	$0, -1$	$p + e^- \to n + \nu$; $Q = [M_X - M_Y]c^2 - B_e$
γ	$0, 0$	nuclear de-excitation

α energy sharing (recoil)

$$K_\alpha = \frac{A-4}{A}Q, \qquad K_{\text{recoil}} = \frac{4}{A}Q$$

β spectrum is continuous (0 to $K_{\max}$) because $\bar{\nu}/\nu$ shares energy. End-point $K_{\max} \approx Q$ (ignore recoil). Neutrino: Pauli, ≈ 0 rest mass in JEE.

★ β^+ needs $Q \geq 1.022\,\text{MeV}$ (two electron masses). EC can occur when β^+ is forbidden.

★ Geiger–Nuttall (advanced): $\log\lambda = a + b\log R_\alpha$ or $\ln\lambda = A - B/\sqrt{Q_\alpha}$ — shorter $T_{1/2} \Leftrightarrow$ higher Q_α.

6.2 Successive decay, mixture, dating ★

$A \xrightarrow{\lambda_A} B \xrightarrow{\lambda_B} C$ (stable), $N_B(0) = 0$:

$$N_B = \frac{\lambda_A N_{A0}}{\lambda_B - \lambda_A}(e^{-\lambda_A t} - e^{-\lambda_B t})$$
$$t_m = \frac{\ln(\lambda_B/\lambda_A)}{\lambda_B - \lambda_A}$$

Secular eq. ($\lambda_B \gg \lambda_A$, parent long-lived): $A_B = A_A$ i.e. $\lambda_B N_B = \lambda_A N_A$.

Transient eq. ($\lambda_B > \lambda_A$): $A_B/A_A = \lambda_B/(\lambda_B - \lambda_A)$.

Two independent species: $A(t) = A_1 e^{-\lambda_1 t} + A_2 e^{-\lambda_2 t}$.

Parallel (branching) modes: $\lambda = \lambda_1 + \lambda_2$, $T_{1/2} = 0.693/\lambda$. Branching ratio $= \lambda_i/\lambda$.

Carbon dating (^{14}C, $T_{1/2} = 5730\,\text{yr}$):

$$t = \frac{1}{\lambda}\ln\frac{A_0}{A} = \frac{T_{1/2}}{0.693}\ln\frac{A_0}{A}$$

A_0: activity of living sample (same species, same mass). Also works with N_0/N.

7 · HIGH-YIELD JEE SHORTCUTS ★

- If λ is halved at fixed energy-intensity I, photon energy $\times 2$, photon number $\times 1/2$, so sat. current halves and $K_{\max}$ changes via Einstein. Many JEE Main keys treat intensity as photon flux, so they keep i independent of ν.
- $eV_s = 12400(1/\lambda - 1/\lambda_0)$ with λ in Å, V_s in volts.
- de Broglie of e^- in Bohr n: $\lambda_n = 2\pi r_n/n = 2\pi a_0 n/Z$.
- Ratio $r_n : v_n : T_n : E_n = n^2 : 1/n : n^3 : 1/n^2$ (fixed Z).
- For H-like, $E \propto Z^2$, $\lambda \propto 1/Z^2$, $v \propto Z$, $r \propto 1/Z$.
- A photon from $n_2 \to n_1$ ionises ground-state H only if $\Delta E \geq 13.6\,\text{eV}$ (Lyman limit / continuum). Lyman α is $10.2\,\text{eV}$ — it excites, it does not ionise.
- To excite H from ground, minimum $\lambda = hc/10.2\,\text{eV} = 1216\,\text{Å}$ (not 911 — that's ionisation).
- X-ray: if V doubled, $\lambda_{\min}$ halves; characteristic λ unchanged (depends on Z).
- If K_α of an element matches Lyman α of H-like ion, use Moseley = Bohr with $Z - 1$.
- Activity in n half-lives: $A = A_0/2^n$. Mass of radioactive atoms $\propto N \propto A$.
- Mixture of α and β emitters: total $A = \sum \lambda_i N_i$; detected particles depend on branching.
- In α decay of rest parent, $p_\alpha = p_{\text{recoil}}$, so $K \propto 1/m$.
- 1 u mass defect $\Leftrightarrow 931.5\,\text{MeV}$. Never use 931 vs 931.5 inconsistently with the paper.
- Nuclear density independent of A; atomic density is not.
- Compton $\Delta\lambda/\lambda$ is large for X / γ, negligible for visible ($\lambda \gg \lambda_C$).
- If stopping potential is V_s at λ, at 2λ it is not $V_s/2$ — use Einstein, not proportion.
- Number of photoelectrons $\propto I \times (\text{area}) \times t/(h\nu)$ times quantum efficiency ≤ 1.

8 · GRAPH & TRAP CHECKLIST

- i–V photoelectric: current zero at $V = -V_s$, then rises to saturation; reversing anode–cathode beyond saturation does not raise $K_{\max}$.
- KE of photoelectrons is a distribution $0 \to K_{\max}$; $K_{\max}$ is for the least-bound (surface) electrons.
- Work function is of the **emitter**, not the collector (unless specified).
- Bohr E_n is **negative**; transitions use $|\Delta E|$. Do not write $E = +13.6/n^2$.
- Rydberg: n_l is the **lower** (smaller) quantum number for emission formula as written.
- X-ray $\lambda_{\min}$ from tube voltage; characteristic λ from target Z. Do not mix.
- Radioactive $A = \lambda N$, not N. A sample can have large N and small A if λ is tiny.
- β^+ Q formula with atomic masses must subtract $2m_e c^2$; β^- must not.
- Half-life of a given nuclide is independent of N_0; activity is not.
- de Broglie: use kinetic energy, not total/relativistic energy, unless the problem is relativistic.

Corrected vs source PDF: Bohr r_n includes ε_0 and $\hbar$; v_n uses $\hbar$ not h; $E_n = -13.6Z^2/n^2$ eV (minus sign, Z); Einstein $K_{\max} = eV_s$; half-life algebra is $N_0/2 = N_0 e^{-\lambda T_{1/2}}$.